Supplementary Material: Physically Implausible Timestamps in CAN Intrusion Detection Benchmarks

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This companion retains procedural and explanatory material moved from the main text. The audit uses the nominal 47-microsecond threshold with a full intermission. The main paper states the 46-bit-time exception, format limitations, corpus-specific eligible gap populations, and the distinction between a timing screen and source identification. For the pseudocode below, $\mathcal{G}_c$ selects consecutive parseable ROAD rows or adjacent normal CTAT rows; attack rows break the latter sequence, negative gaps are excluded and zero gaps retained. Same-identifier intervals never cross source files. Integer microseconds and exact decimal seconds are compared in matching units.

S1. Physical-floor construction

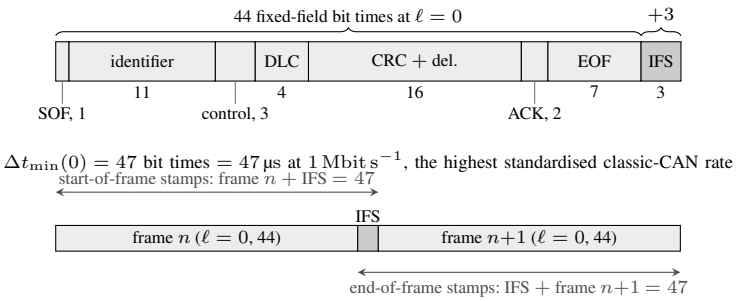


Fig. S1. Where the bound of the nominal floor $\Delta t_{\min} = (47 + 8\ell)/R$ comes from, at $\ell = 0$. The frame's seven fixed fields sum to 44 bit times and the interframe space adds three: a nominal 47-bit-time lower bound, $47\,\mu s$ at the highest standardised rate of $1\,\mathrm{Mbit\,s^{-1}}$. The lower panel shows both stamping conventions on a common bit-time scale: at $\ell = 0$ start-of-frame and end-of-frame stamping span the same 47 bit times, so the bound does not depend on where the logger stamps. Bit stuffing is excluded because it can only lengthen a frame.

S2. AUDIT PSEUDOCODE

Algorithm 1: Exact-arithmetic audit specification for ROAD and can-train-and-test

Input: ROAD or can-train-and-test capture files $\mathcal{C}$
Output: per-file verdict, $\rho_<$, epoch and span flags, per-identifier cross-check

$\Delta t_{\min} \leftarrow 47\,\mu\text{s}$; // $\Delta t_{\min} = (47 + 8\ell)/R$ at $R = 1\,\text{Mbit/s}$, $\ell = 0$; 47 or 0.000047 in file units
foreach *file* $c \in \mathcal{C}$ **do**
  parse timestamps exactly as $\hat{t}_1 \dots \hat{t}_n$: integer microseconds where the log has fixed precision, exact decimals otherwise ; // exact timestamp-string parsing; can-train-and-test takes the second branch
  $\tilde{g} \leftarrow$ successive differences of $\hat{t}$, exact in that representation;
  select eligible gap indices $\mathcal{G}_c$ using the corpus-specific rule above;
  $\rho_< \leftarrow |\{i \in \mathcal{G}_c : \tilde{g}_i < \Delta t_{\min}\}|/|\mathcal{G}_c|$;
  epoch $\leftarrow \mathbb{1}[\hat{t}_1$ is a whole number of seconds$]$;
  span $\leftarrow \hat{t}_n - \hat{t}_1$, rescaled to seconds; compare with declared duration;
  foreach *identifier* $a \in \mathcal{A}(c)$ **do**
    $\tilde{\nu}_a \leftarrow$ median of positive same-identifier intervals ; // label-free; not the ν_a of the main paper's benign-training nominal period
    flag a iff $\tilde{\nu}_a < \Delta t_{\min}$;
  verdict $\leftarrow$ TIMING-REFERENCE IMPLAUSIBLE **if** $\rho_< > 0.01$ **else** PASSES THIS SCREEN ; // empirical screen, not source identification; the main paper's conclusion
return *verdicts*;

S3. DAMAGE TAXONOMY

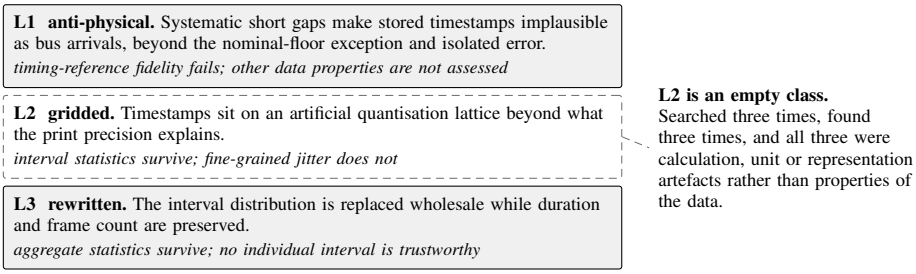


Fig. S2. The three damage levels of the main paper's taxonomy. The italic line in each box states what that level leaves usable and what it destroys; the three differ, which is why "synthetic" alone does not tell a practitioner which uses are affected. L2 is drawn dashed and empty because no genuine instance appears in the four corpora audited here—a result, not an unfinished search.

S4. Published statements and measurements

TABLE S1
DOCUMENTED TIMESTAMP HANDLING IN ROAD AGAINST MEASUREMENT. THE LAST ROW IS THE AUTHORS' OWN METHODOLOGICAL POSITION, APPLIED TO A DIFFERENT CORPUS.

Published statement	Measurement
"Absolute timestamps are shifted uniformly by a scalar" [1, 2]	A constant shift preserves all gaps. Observed: sub-floor gaps and integer-second starts in 45/45 files.
Timestamps "are reported with a precision of 1 µs", while the capture hardware "only guarantees an accuracy of 100 µs" [1, 2]	Stated accuracy alone does not explain sub-floor gaps at a 1 µs median; print precision is not a capture-time error model.
Masquerade captures are "slightly modified in post-processing" by "removing the legitimate target ID frames preceding each injected frame" [1]	Frame removal merges gaps, creates none; ambient captures, uncovered by any modification statement, carry the same signature.
"Adjusting timestamps in post-processing alters data and diminishes fidelity in a critical way" [1]	The authors' own criterion, stated when assessing a different corpus.

S5. Current and historical sensitivity evidence

The current complete run contains 58 evaluation files, 141 configurations per file and 8178 result rows. The corrected SynCAN reader retains the first normal frame of each of three headerless training files. Stored metrics change in 610 SynCAN rows relative to the historical result; unknown historical code, input and cache identities prevent attribution solely to the three restored rows. The new run replaces those sensitivity estimates; it does not retrospectively certify the historical runtime. The CTAT and ROAD groups contain 40 and 13 evaluation files, and SynCAN contributes five attack test files; its separate baseline also includes the normal test file. The main paper reports each numeric estimand with its population and precision.

References

[1] M. E. Verma, R. A. Bridges, M. D. Iannacone, S. C. Hollifield, P. Moriano, S. C. Hespeler, B. Kay, and F. L. Combs, "A comprehensive guide to CAN IDS data and introduction of the ROAD dataset," *PLOS ONE*, vol. 19, no. 1, p. e0296879, 2024.

[2] M. E. Verma, R. A. Bridges, M. D. Iannacone, S. C. Hollifield, P. Moriano, B. Kay, S. Hespeler, and F. L. Combs, "Real ORNL automotive dynamometer (ROAD) CAN intrusion dataset," Zenodo, 2020, dataset record; its technical description carries the timestamp-handling statements quoted in the text.